

Assignment Day 3

Page No.

Date

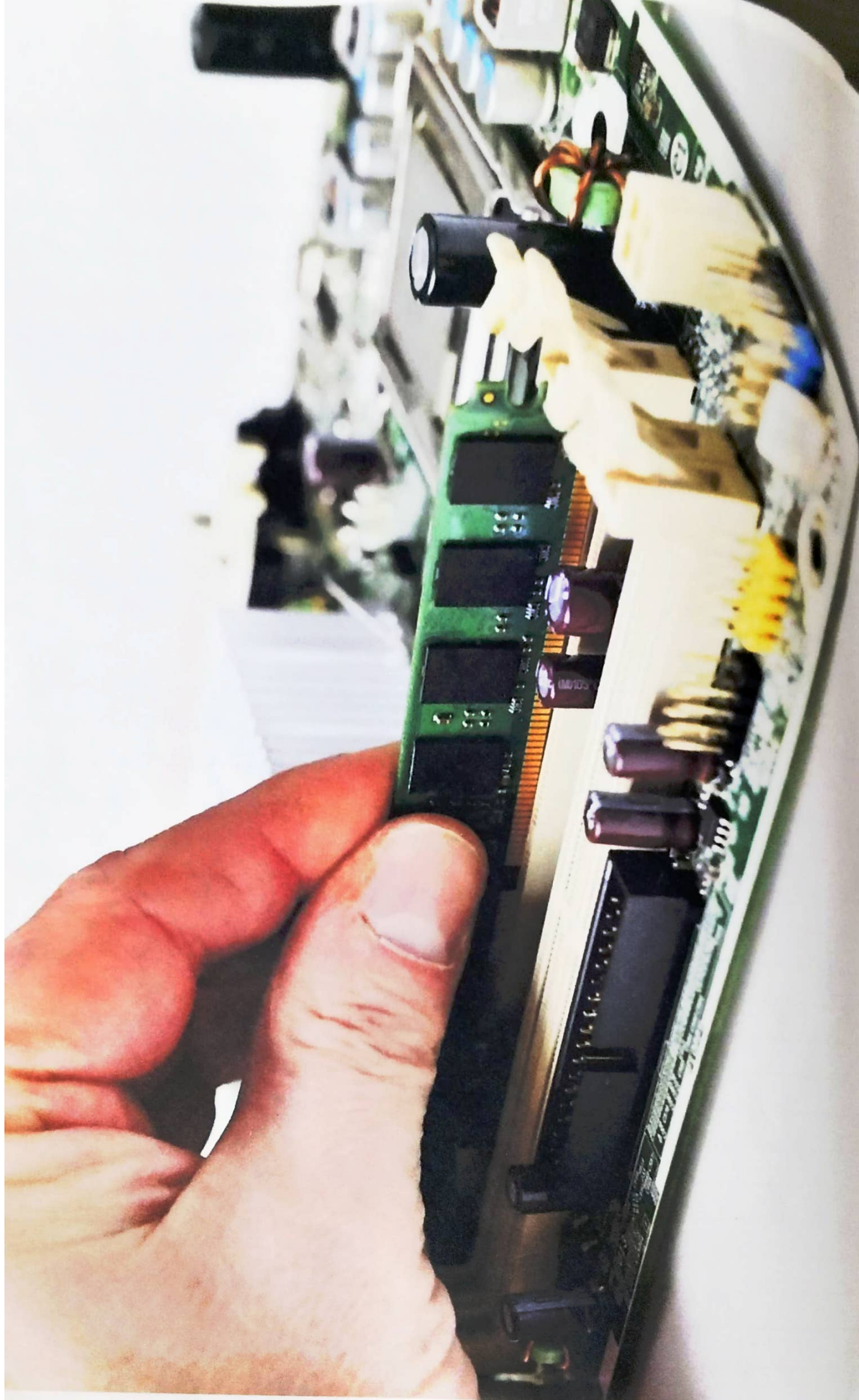
Day 3.

1. Create a comparison table listing the main differences between DDR3, DDR4 and DDR5 RAM. Including Speed, voltage, and typical use cases.

Features	DDR3 (Old Standard)	DDR4 (Current Standard)	DDR5 (New Standard)
Speed	800 to 2133 MT/s	2133 to 3200 MT/s	4800 to 8400+ MT/s
Voltage	1.5V (Less Power)	1.2V (Balance)	1.1V (Most Power)
Max Memory Size	8 GB	32 GB to 64 GB	128 GB or more
Typical Use Cases	Basic office PCs, older laptop, legacy systems	General computing, budget gaming and everyday office work.	High-end gaming, 4K video editing, 3D rendering and AI workloads.

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2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

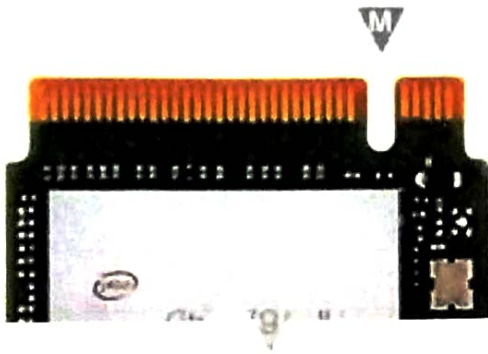


3. List three key differences between HDD, SSD and NVMe storage types and for each name one app or use case (from apps you use daily) that would benefit most from that storage type.

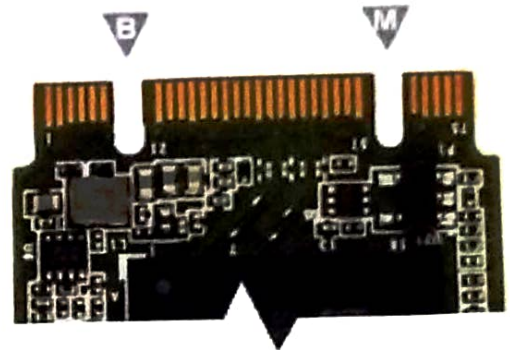
Features	HDD (Hard disk drive)	SSD (Solid state drive)	NVMe (Non-volatile memory express)
Technology & speed	HDD use spinning mechanical disks. Speed = 100-160 MB/s	SSD use Flash memory chips with no moving parts. Speed = 500-550 MB/s	NVMe use ultra fast flash memory plugged directly into the motherboard's PCIe slots. Speed = 3500-7000+ MB/s
Price per GB	HDD's price is cheapest	SSD's price is moderately price.	NVMe price are very expensive.
Durability & Reliability	HDD can fail if dropped because of fragile mechanical parts	SSD have zero moving parts, making them silent & durable & shock resistant.	NVMe also have zero moving parts, making them silent, durable & shock resistant.
Ideal app / daily use case	Google Drive / Dropbox / Local sync (or storing large photo archives)	Google Chrome / Microsoft Edge	Adobe Premiere Pro (or heavy modern games like Cyberpunk 2077)
Benefit	Offer massive cheap storage ^{where} for transferring doesn't matter much for background backups	Flash memory instantly opens web browsers and system app, eliminating startup delays	massive bandwidth lets you scroll through 4K video footage or load huge gaming worlds instantly without stuttering.

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4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

M KEY

(M.2 NVMe)

B+M KEY

(M.2 SATA)